

HelixQA Go orchestrator — dependency submodules wired + build PROVEN (2026-06-23)

Revision: 1 **Last modified:** 2026-06-23T16:30:00Z **Operator directive:** add the HelixQA submodule dependencies into the proper location per the constitution (§11.4.28(C)/§11.4.31/§11.4.36) so the Go orchestrator builds. Verdict: **DONE** — **real build proof.**

What was added (7 dependency submodules at submodules/<name>, §11.4.28(C))

submodule path	repo	HEAD	content
submodules/ containers	vasic-digital/ containers	e635ad8	✓
submodules/ doc_processor	vasic-digital/ DocProcessor	b8c71de	✓
submodules/ llm_orchestrator	vasic-digital/ LLMOrchestrator	a484f7d	✓
submodules/ llm_provider	vasic-digital/ LLMProvider	8a4ca59	✓
submodules/ llms_verifier	vasic-digital/ LLMsVerifier	09bcaec	✓ (shallow — see note)
submodules/ security	vasic-digital/ security	b69c7d9	✓
submodules/ vision_engine	vasic-digital/ VisionEngine	0bf75ee	✓

These match helixqa’s go.mod replace paths exactly (../<name> from submodules/helixqa → submodules/<name>).

Path-mismatch bridge (§11.4.29 inconsistency)

submodules/llm_orchestrator/go.mod references ../LLMProvider (CamelCase) while helixqa references ../llm_provider (snake_case) — the same repo at two conventions. Bridged with a consumer-side relative symlink submodules/LLMProvider → llm_provider (parent-level, NOT injected into any

submodule per §11.4.28(B); not a go.work, to avoid affecting the ota-* builds). The llmsverifier module lives at the subpath submodules/llms_verifier/llm-verifier/ (matching helixqa's ../llms_verifier/llm-verifier replace).

Build proof (rock-solid, §11.4.6 — real exit codes)

```
cd submodules/helixqa
go build ./...                → BUILD_EXIT=0    (clean)
go build -o helixqa ./cmd/helixqa → CMD_EXIT=0    (binary
29,940,418 bytes - links)
go vet ./pkg/testbank ./pkg/orchestrator → clean
```

The HelixQA Go orchestrator (cmd/helixqa: run|list|report|autonomous|http|...) now compiles + links against the OTA project's submodule layout. Per the earlier WIRING_PLAN, the OTA bank format already matches; the remaining step to drive it vs the live OTA system is a thin DeviceExec host-exec adapter for the dispatches_to challenges (separate follow-up).

Honest notes (§11.4.6)

- **LLMsVerifier clone:** the full GitHub clone fails repeatedly with fetch-pack (a deep-history object-transfer error, not LFS, not empty-repo — the repo is 388 KB, pushed today). A --depth 1 shallow clone succeeds and provides the working tree (1316 files) the build needs. It is registered as a submodule; a future git submodule absorbgitdirs + full-history re-fetch (once the upstream pack issue is resolved) is a tracked follow-up.
- **install_upstreams (§11.4.36):** run per added submodule where an upstreams/ recipe dir is present.
- This wires the orchestrator to BUILD; running a full HelixQA autonomous session vs the live OTA system is the next step (needs the DeviceExec adapter + thinker).